Derek Eubanks PE, HFDP, HBDP, CHD, CPD

Atlanta, GA 30329 | Hawthorne, CA 90250 | 470.530.4898 | Derek.Eubanks89@gmail.com | LinkedIn | GitHub

Licensed Mechanical Engineer (PE – CA, GA, TN) with 13 years of experience designing and leading critical HVAC, hydronic, and plumbing systems for mission-critical facilities, including hyperscale data centers, healthcare environments, cleanrooms, and advanced manufacturing. Proven expertise in high-performance airside and waterside system architecture, variable-primary-flow hydronic plants, heat pump and energy recovery systems, and complex ventilation and life-safety-driven designs. Experienced in developing constructible, operations-focused solutions supported by detailed load analysis, controls schematics, and sequences of operation. Currently a Master of Science in Computer Science (Machine Learning) graduate student at Georgia Tech (OMSCS), bringing a systems-level, data and machine learning–driven approach to engineering design and optimization.

Professional Certifications & Licenses:

- Professional Engineer (PE Licenses): California (#42960), Georgia (#PE042696), Tennessee (#130418)
- MyNCEES PE-verified engineering employment history since 2013 – ability to get licensed via comity in all 50 states
- ASHRAE Healthcare Facility Design Professional (HFDP)
- ASHRAE Certified HVAC Designer (CHD)
- ASHRAE High-Performance Building Design Professional (HBDP)
- ASPE Certified in Plumbing Design (CPD)
- Autodesk Certified Professional in Revit for Mechanical Design

Education:

- Bachelor of Science in Mechanical Engineering (BSME) | Georgia Institute of Technology (Aug 2007 - May 2013)
  - Dean's List (2) semesters; Merit list (2) semesters
  - Math GPA: 3.68/4.0 (7 courses); Computer Science GPA: 3.5/4.0 (2 courses)
  - HOPE scholarship & Federal PELL Grant recipient for maintaining >= 3.0 GPA
- Master of Science in Computer Science (MSCS) | Georgia Institute of Technology (Aug 2025 - Dec 2028)
  - Specialization in Machine Learning
  - Completed: CS 6603 (A), CSE 6242 (A)
  - In Progress: CS 6310 + CS 7646 – Summer 2026
  - 4.0/4.0 GPA

Graduate Coursework & Research Projects – Applied Machine Learning:

- Algorithmic Fairness Analysis in Employee Promotion Models – AI, Ethics, and Society (CS 6603-Fall 2025):
  - Applied machine learning (logistic regression, random forest) to analyze 54k+ records and quantify bias using SPD/EOD metrics
  - Implemented fairness metrics including Statistical Parity Difference (SPD) and Equal Opportunity Difference (EOD) to quantify disparities across protected groups (gender and age).
  - Trained and evaluated Logistic Regression models on both original and bias-mitigated datasets using a reweighing algorithm, demonstrating fairness improvements without degrading predictive performance.
- Airline Delay Root Cause Explorer – Data & Visual Analytics (CSE 6242-Spring 2026):
  - Led Python-based analysis of large-scale airline operations datasets (400k–800k+ records; BTS + NOAA), developing machine learning models to identify systemic drivers of flight delays.
  - Built end-to-end ML pipeline (data cleaning, feature engineering, preprocessing, modeling, and evaluation) using Random Forest, Logistic Regression, and Gradient Boosting, achieving ROC-AUC ~0.70 and optimizing recall/F1 through threshold tuning.
  - Performed model comparison and error analysis to quantify tradeoffs between accuracy, precision, and recall, demonstrating that model selection and thresholding significantly impact delay detection performance.
  - Identified key drivers of delay using feature importance and model interpretation, showing that scheduled departure/arrival timing, weather (precipitation, wind), and airport operational activity are primary contributors.
  - Contributing to development of an interactive analytics platform (D3/Tableau) enabling exploration of delay patterns across U.S. airports, seasons, and operating conditions.

Technical Skills:

- Programming: Python, VBA, C#; Visualization: D3.js, Tableau, Matplotlib, Seaborn.

- Data & Machine Learning: Scikit-learn (Logistic Regression, Random Forest, Decision Trees, Model Evaluation), Statistical modeling and predictive analytics, Bias and fairness evaluation (SPD, EOD), Pandas, NumPy, SciPy.
- Engineering Software: Revit, Trace 3D Plus, Carrier HAP, Bluebeam, Microsoft Office Suite.
- HVAC Systems: Experience with heat recovery CHW VAV, DX and WSHP systems, VRF heat pump/recovery, and active/passive desiccant DOAS ERUs.
- Plumbing Design: Design of domestic water, sanitary, vent, storm, medical gas (N2, Vacuum, CO2, Med air, O2, WAGD, NO2), natural gas; Advanced Revit modeling integration in both 2d and 3d.
- Controls & Sequences: Create airside/waterside Revit control schematics, I/O points list, and detailed sequences of operation for each component.
- Leadership & QA/QC: Lead discipline coordination meetings, perform QA/QC reviews, write proposals, and mentor and utilize junior engineers on projects to help them grow.
- Equipment Selection: Sizing and selection of air/water-cooled chillers, fluid coolers, DX/CHW AHUs, CRAHs, DOAS ERUs, fans, pumps, FCUs, RTUs, VAVs, PIUs, and air valves.
- Codes & Standards: Applied knowledge of ASHRAE 170, 90.1, 62.1, 55, 15, 100, 105, 228, IMC, IPC, IECC, NFPA 90A/99, and FGI Guidelines.

Revit MEP Experience:

- Medium/low pressure air distribution (SA, RA, EA, OA) layouts for large hospitals, data centers, college campus buildings, and various office buildings – from the mechanical room AHU to the terminal units, to the space diffusers and back via the ducted/plenum return system – I have experience modeling the entire air distribution in Revit.
- Experience with hydronic chilled/hot water piping systems – auto sizing piping based on flow, velocity, pressure drop.
- OR, AII, PE, patient room, Class III Imaging, pharmacy, lab, and other ASHRAE 170 and 62.1 space HVAC design.
- VAV, PIU, and supply/return/exhaust air valve terminal unit placement, sizing, selection, and scheduling.
- Detailed experience with mechanical and plumbing schedules and filtering/sorting with conditional statements.
- Detailed knowledge of pressure loss reports and using the duct/pipe sizing tool effectively with appropriate families.
- Mechanical Room layouts including AHUs, Pumps, DOAS, ductwork, piping, air separator, expansion tank, buffer tanks.
- Modeled sanitary/waste, vent, domestic water, rain leader, medical gas 2D and 3D riser diagrams per code.
- Tier III data center chiller yard with 40+ air-cooled chillers and chilled water piping coordinated with other disciplines.
- Electrical Room/CRAH Gallery/Data Hall CRAH layout with chilled water piping and condensate piping modeled in 3D.
- Developed custom Revit mechanical and plumbing families that include 3D extrusions/sweeps/lofts, pipe/duct connectors and their involvement in duct/pipe systems in the Revit model; instance and type parameters and conditional statements, reference planes/lines, reference level, isometric, and section view modifications; different types of instance/type parameters for flow rate, pressure drop, fixture units, dimensions, yes/no, etc.
- Healthcare medical gas system design and modeling per NFPA and BeaconMedaes including Zone Valve Boxes, alarm panels, piping, valves, outlet terminal families, pressure sensors. The outlet terminal families contain flow and diversity.
- Experience with customized view templates, view phasing and phase filtering, view creation and placement on sheets.

Professional Engineering Employment History (2013-Present):

- Sr. Mechanical Engineer (Critical Infrastructure) – SpaceX | July 2025 – Jan 2026; Hawthorne, CA – On site
  - Lead design, analysis, and optimization of fluid and HVAC systems for world-class manufacturing facilities supporting SpaceX's multiplanetary mission.
  - Perform detailed load calculations in Trace 3D Plus to size and optimize HVAC and hydronic systems.
  - Engineer and coordinate heat pump chiller hydronic plant systems in Revit (piping layouts, equipment selection, integration, commissioning readiness).
  - Full 3D modeling: Build complete mechanical, hydronic, and HVAC models in Revit, placing equipment with maintenance/access clearances, service envelopes, and code/safety offsets; use intelligent (parametric, data-rich) Revit families to drive schedules, tagging, and coordinated shop-ready deliverables.
  - Develop controls schematics and detailed Sequences of Operation (SOO) for chilled-water plants, airside systems, and ducted supply/return/exhaust air distribution.
  - Dust collection optimization: Capture VFD frequency, airflow, and duct static pressure to develop system curves; set minimum fan speed to maintain minimum duct velocity for dust suspension across open/closed damper positions; establish vacuum setpoints and tune damper control.
  - Oversee hydronic pumps, compressors, heat exchangers, dust collection systems, and compressed air distribution.
  - Utilize Revit, CAD, and performance simulation tools to deliver efficient, maintainable, and high-performance designs.

- Collaborate with architects, mission operations, and manufacturing to deliver reliable, high-capacity infrastructure solutions.

- Senior Associate | Senior Mechanical Engineer – [Syska Hennessy Group] | Jan 2024 – March 2025; Atlanta, GA - On site
  - Led full mechanical design for a 48 MW hyperscale data center, ensuring compliance with Uptime Institute standards for N+1 redundancy and fault-tolerant system architecture.
  - Oversaw HVAC and chilled water system layout across all critical zones, including data halls, UPS and battery backup rooms, electrical galleries, and mechanical yards.
  - Modeled and coordinated complex 3D mechanical spaces in Revit, including hydronic CRAH galleries, dedicated mechanical rooms, and chilled water infrastructure, ensuring clash-free integration with other disciplines.
  - Produced complete mechanical construction documents in Revit, including airside systems (SA/RA/OA/EA ductwork) and hydronic systems (CHWS/CHWR, condensate) supporting CRAHs, modular air-cooled chillers, and DOAS ERU units.
  - Executed space-level cooling load calculations using Trane Trace 3D Plus and Excel for high-density white space and supporting infrastructure, aligning capacities with continuous operation requirements.
  - Coordinated directly with OEMs for selection of equipment including modular chillers, DOAS ERU systems, expansion tanks, and VFD-driven primary pumps.
  - Directed a multidisciplinary mechanical team, delegating design of CRAH unit piping, CHW loop schematics, CFD-informed airflow strategies, isometric riser diagrams, and construction-ready mechanical schedules.
  - Implemented N+1 redundancy across the chilled water plant, including dual CRAH supply headers, looped CHW distribution, and valve isolation zones, ensuring systems could remain operational during maintenance/failure.

- Senior Mechanical Engineer – [Jacobs] | Oct 2022 – Dec 2023; Atlanta, GA - Fully-remote
  - Executed mechanical HVAC and plumbing design in Revit for large-scale Tesla and Rivian manufacturing facilities.
  - Applied client-provided, space-specific environmental criteria to guide design and coordination efforts.
  - Coordinated HVAC system selection to meet strict temperature and relative humidity tolerances.
  - Deployed heat pumps as the primary heat source for AHU coils, desiccant wheels, and domestic hot water HXs.
  - Modeled and coordinated dozens of 40,000+ CFM chilled water AHUs in Revit, following detailed load analysis.
  - Designed a hydronic heating system to raise a specialty material storage room from ambient to 140°F within timeframe.
  - Performed iterative load calculations using Trane Trace 700 to reflect evolving Revit model and envelope conditions.
  - Delegated routine calculations to junior engineers, including external static pressure and hydronic pump head estimates.
  - Designed and modeled a VAV DX heat pump HVAC system for support areas on the Tesla project in Revit.

- Senior Mechanical Engineer – [Mazzetti] | Apr 2020 – May 2023; Atlanta, GA - Fully-remote
  - Developed Trane Trace 3D+ compatible Hospital Air Balance VBA Excel spreadsheet for detailed space and AHU analysis.
    - GitHub: https://github.com/dwayne902642323/Trane-Trace-3D-Plus-Air-Balance-VBA-Spreadsheet
  - Designed and modeled in Revit: HVAC, plumbing, and medical gas systems for healthcare facilities in CA and GA.
  - Performed heating/cooling load calculations using Trane Trace 700 and Trace 3D Plus and Excel.
  - Employed series heat recovery coils in CHW AHUs that serve ORs to reduce unnecessary reheat and meet energy code.
  - Performed detailed calculations such as determining the leaving DOAS ERU supply air dew point temp, via passive desiccant dehumidification, required to offset the variable and peak latent space loads from all moisture sources.
  - Additional calculations included dynamic total head loss for hydronic pumps and critical run external static pressure loss for supply/return fans in large AHUs and exhaust fan systems.
  - In Revit, I laid out several mechanical rooms with CHW/HW AHUs, DOAS ERUs and associated ductwork/piping.
  - OR suites with supply/return air valves, PE and AII rooms with positive and negative differential pressure control respectively via supply/exhaust air valves.
  - Designed and modeled in Revit, the stairwell pressurization fan and duct system for the UC Davis Hospital project.
  - Developed detailed air/water side controls schematics, points list, and sequences of operation for healthcare HVAC systems in Revit drafting views to include in mechanical drawing package.

- Mechanical Discipline Director – [Gray AES] | Aug 2018 – Mar 2020; Peachtree Corners, GA - On-site
  - Delegated project assignments to mechanical engineers and designers.
  - Develop a VBA-driven Excel spreadsheet to track all project fee information for the company.
  - Lead weekly mechanical discipline meetings.
  - Engineer of Record on mechanical/plumbing projects.
  - Designed HVAC and plumbing systems for Delta, Pratt & Whitney, and other industrial clients.
  - Developed a Data Center Technical Report for mechanical chilled water system study impacts on expansion projects.

- Conducted performance reviews with mechanical engineers and designers.
- Developed proposals for new projects – which led to billable projects.
- Project manager and EOR for WellStar AMC Class III Imaging Room HVAC, plumbing, and medical gas project.

- Mechanical Engineer II – [Pond & Company] | Apr 2016 – Aug 2018; Peachtree Corners, GA - On-site
  - Designed HVAC and plumbing systems for government and higher education projects.
  - Performed load calculations using Carrier HAP and Trane Trace 700.
  - Presented to the mechanical team on various topics including ASHRAE 62.1 guidelines.
  - Designed, calculated, and modeled in Revit, the mechanical and plumbing systems for the Central Chiller Plant for the UNG project. The sanitary, vent, domestic water, HVAC air distribution and mechanical chilled/hot water piping.
  - Developed an ASHRAE Excel VBA Psychrometric calculation spreadsheet with inputs/outputs and precise results.
  - Utilized UFC codes for HVAC/plumbing design and applied necessary code aspects to Federal projects.
  - Performed energy models via 90.1 in Trane Trace 700 for Federal projects. Building rotation method was applied.
  - Designed the HVAC systems for small data center rooms that support the building functions for federal projects.

- Mechanical Designer – [Mazzetti] | Apr 2013 – Apr 2016; Atlanta, GA - On-site
  - Designed HVAC and plumbing systems for healthcare projects in Atlanta, GA.
  - Designed, sized, and modeled in Revit the domestic water, sanitary, vent, and medical gas piping for the CTCA project.
  - Performed HVAC cooling/heating load calculations using Elite HVAC software.
  - Psychrometric calculations such as mixed air and required LAT for cooling coils. Hydronic pump head loss calculations.
  - Developed and created Revit plumbing families for company use on projects. Extensive research and development.
  - Modified and compiled community-sourced C# code in Visual Studio to automate domestic water pipe sizing in Revit using modified Hunter's Curve flow conversions per IPC.
  - Presented at ASPE for Revit plumbing family development.

Revit API (C#) Programming – Engineering Software Development:

1. [Revit C# Add-in – User Plumbing Fixture Flow Server (Predominantly Flush Valve Systems)]
   - Inspiration: Sizing domestic water piping manually for large models was time-consuming and prone to errors, especially with flush-valve fixture units.
   - Approach: Developed custom families in Revit that automatically convert fixture units to volumetric flow rates. Used fluid mechanics equations (friction factor, velocity limits, pressure drop) coded in C# to auto-adjust pipe diameters.
   - Achievement: Eliminated extensive manual calculation. Revit families handle complex parametric conditions, ensuring compliance with code-based velocity or friction constraints. Presented at ASPE, validated by CPDs and PEs.
   - Result: Drastically reduced design hours, improved uniformity in mechanical deliverables.
   - Developed a custom Revit C# API add-in that overrides Revit's default fixture unit (FU) to flow rate logic for plumbing systems. The add-in expands flush valve flow conversion accuracy for fixtures under 5 FU by implementing IPC/ASPE interpolation tables and custom Hunter's Curve logic.
   - Registered via the PlumbingFixtureFlowServer interface, the tool ensures correct GPM values are applied to piping systems below 15 GPM, enabling precise pipe sizing and velocity and pressure drop checks directly within the Revit model. Significantly enhances design automation and standards compliance for domestic water systems.

MEP Engineering Projects:

1. [UNG Campus Chilled Water Plant and Technology Building HVAC/Plumbing Design in Revit] (Pond & Company)
   - Led mechanical and plumbing design for the University of North Georgia's campus central plant facility, providing chilled water to multiple buildings via a campus-wide loop. Modeled and engineered all major systems in Revit, including the chilled water plant and HVAC/plumbing infrastructure within the central plant building. The plant featured two series counterflow chillers, double-suction chilled and condenser water pumps, a closed-loop fluid cooler, waterside economizer, air separator, and expansion tanks.
   - $40 million construction cost – This 103,000 sqft facility became the major "anchor" load for the central plant. The mechanical/plumbing design had to account for massive swings in occupancy (from empty to 4,000+ people), requiring the variable primary Flow (VPF) system to be extremely responsive. $10 million for the central plant equipment and distribution piping alone.
   - 3,000-ton total plant design using series counterflow chillers operating at a larger delta T (compared to parallel) to increase efficiency and realize a 25-30% reduction in annual pumping energy compared to a standard primary-secondary configuration. Each building had its own secondary pump and loop serving building air handling units.

- In series counterflow setup, the chillers were paired (Chiller 1A and 1B) where one handles the "high lift" and the other handles the "low lift" to achieve a combined 1,500 tons per pair – each contributing to lower the water temp at the same flow rate – yielding a high total pressure drop across the two chillers in series. Bypass piping/valves were the solution, by combining the right combination of piping and controls valves, either chiller could be bypassed. This is what made the variable primary loop so effective as buildings came “online” and additional capacity was required.
- Performed detailed load calculations for the central plant and connected campus buildings to determine chiller capacity, tower performance, and pump sizing. Designed the building’s HVAC system around a variable air volume (VAV) air handling unit with hot water reheat, supplemented by an energy recovery unit (ERU) with an enthalpy wheel. The ERU captured energy from toilet exhaust and building relief (return) air to precondition outside air and maintain balanced pressurization. Total energy wheels were used with a typical total efficiency around 80% in GA.
- Plumbing systems were modeled and sized using custom Revit families with Hunter’s curve-based fixture unit (FU) to flow conversions. Domestic water, sanitary, and vent systems were fully designed per the International Plumbing Code (IPC), with isometric diagrams created for clear coordination and construction sequencing.
- Developed the central plant control sequence, points list, and schematics to govern chiller staging, pump sequencing, and waterside economizer logic. Designed the system for energy efficiency, operational flexibility, and adherence to ASHRAE 90.1 and IPC requirements. Coordinated closely with structural, architectural, and electrical teams to resolve spatial constraints and maintain code compliance in all mechanical and plumbing systems.

2. University of California Davis Hospital OR Suite HVAC Design (Mazzetti)
   - Served as the lead mechanical designer for the operating room (OR) suite within a new 14-story hospital tower and $3.7 billion dollar investment - at the University of California, Davis. 334 beds and 12 hybrid operating rooms. The scope included full mechanical design, load calculations, and 3D Revit modeling of the operating rooms served by a dedicated mechanical penthouse floor. The design emphasized strict adherence to ASHRAE 170, FGI Guidelines, and California healthcare code requirements, with a focus on infection control, airflow reliability, and energy efficiency.
   - Critical pressurization control: Each OR was supplied with 25 air changes per hour via variable volume air valves. Modulating exhaust valves, controlled by differential pressure sensors, maintained a stable +0.02” w.g. pressure offset to adjacent spaces, ensuring directional airflow and contamination control.
   - Thermal and humidity control: ORs were maintained at 65°F and 50% RH, with chilled water systems carefully sized to handle the space’s high sensible load. Series heat recovery coils were utilized in the AHUs to reduce unnecessary reheat during low load conditions. Design accounted for redundancy and control required in surgical settings.
   - Reheat strategy optimization: Reheat was centralized at the AHU level (rather than terminal-level), reducing complexity and maintenance burden in the ceiling plenum. This strategy ensured faster commissioning and improved controllability.
   - DOAS ERU integration: Dedicated outside air systems delivered preconditioned, dehumidified low dewpoint outside air directly to each AHU. The DOAS ERU units handled both latent loads and outside air requirements, allowing the chilled water AHUs to address only the sensible load — improving system efficiency and dehumidification reliability.
   - Utilized Revit to model all AHUs, air distribution systems, duct routing, and control devices in full 3D. This coordination supported early clash detection and streamlined collaboration with architecture and structural disciplines. This project exemplifies high-performance mechanical design for critical healthcare spaces.

3. Tier III Hyperscale Data Center (48 MW) (Syska Hennessy Group)
   - Served as the lead mechanical engineer for a hyperscale data center project, managing the full mechanical design scope from early coordination through permit-level documentation and construction support. Directed a team of mechanical engineers and designers, overseeing detailed development of all HVAC, chilled water, and ventilation systems in compliance with ASHRAE and local code requirements.
   - Led the mechanical design of multiple infrastructure zones including mechanical rooms, data halls, CRAH galleries, and chiller yards. Using Revit and custom Excel tools, I advanced the model to design development level — detailing all chilled water distribution, airside systems, and critical infrastructure in 3D to ensure constructability and clash-free integration. Oversaw CHW system design, including air-cooled chiller banks, mechanical pump rooms, and CRAH unit connections — designed using ASHRAE 90.1 velocity and pressure drop limits for energy efficiency and long-term performance. The chilled water system was distributed to the CRAH units using (3) separate pump rooms piped in parallel and the main chilled water piping was downsized and routed such to avoid non-typically large pipe sizes.
   - Performed load calculations for data halls, electrical rooms, and mechanical spaces using custom spreadsheet logic tailored to sensible heat-dominant zones. Supported hydronic network optimization by collaborating on a full-system PipeFlow model to evaluate chilled water pressure drop and identify pump head reduction strategies.

- Engineered ventilation and pressurization systems for code-required minimum OA delivery using DOAS units, applying ASHRAE 62.1 principles and IMC standards. Selected and configured equipment to maintain neutral pressure in electrical and mechanical rooms while limiting infiltration to data halls.
- Modeled all HVAC ductwork and chilled water piping systems in Revit, maintaining full coordination with structural and electrical disciplines. Issued permit-ready construction documents and continued involvement through construction administration, reviewing submittals, answering RFIs, and confirming adherence to the design intent and applicable codes throughout all phases.

4. CTCA Expansion Projects – HVAC, Plumbing, and Medical Gas Design (Mazzetti)
    - I designed and modeled the HVAC, sanitary, vent, domestic water, and medical gas piping in Revit for the Cancer Treatment Centers of America Expansion project, which included a significant building addition to the Newnan County Hospital. As part of this effort, I first captured existing conditions in Revit based on 3D CAD drawings, ensuring all vent, sanitary, domestic water, and medical gas piping within the scope of work was accurately depicted.
    - 100,000-square-foot, $90 million, expansion phase including a new four-story inpatient wing and a radiation oncology center. 25 new inpatient rooms, each requiring individual climate control and medical gas integration.
    - Installation of advanced HEPA filtration systems and specialized exhaust for the new radiation oncology and surgical suites. Expansion of the central utility plant to support the increased cooling and heating load of the new wing.
    - Once the existing infrastructure was established, I conducted detailed design calculations to properly size all piping according to demand, diversity factors, and system pressure losses. For the domestic water networks, I utilized proprietary plumbing families in Revit—along with a community-sourced C# Revit add-in—to achieve one-step automated sizing, substantially accelerating the design workflow.
    - On the HVAC side, I performed load calculations in Elite HVAC and later exported the results to Excel for additional analysis. This extra step enabled a more comprehensive evaluation of how each space served by the air handling unit (AHU) would respond to both sensible and latent loads. From there, I compared total air change–driven airflow to sensible load–driven airflow, also factoring in outside air change rates to ensure minimum ventilation requirements were met for terminal units and air valves. This final dataset provided a foundation for the overall air balance, which was then integrated back into Revit to develop detailed air balance diagrams.
    - To further refine system performance, I conducted psychrometric analyses for the AHU cooling coil and determined supply/return fan static pressures as well as terminal unit reheat capacities. These calculations were essential for maintaining the correct temperature and humidity levels—particularly critical in healthcare environments where patient comfort and code compliance are paramount.

5. Wellstar AMC - Class III Imaging Room - HVAC, Plumbing, Med Gas Design in Revit (Gray AES)
    - Served as Project Manager, Mechanical Engineer of Record (EOR), and lead HVAC/plumbing designer for a healthcare project involving the design of a Class III imaging room, classified as an operating room per ASHRAE 170. Responsible for full mechanical, plumbing, and medical gas design—producing permit-ready construction documents and managing project timelines, design coordination, and interdisciplinary collaboration.
    - Key responsibilities included psychrometric analysis, air change calculations, and full air balance planning to ensure ASHRAE 170 compliance and appropriate space pressurization. Designed HVAC systems to maintain 25 ACH and a +0.02” w.g. pressure differential in the Class III imaging room, which included an adjacent fan coil unit (FCU) room, overlook room, and administrative spaces.
    - Selected a chilled water FCU with integral humidifier for precise temperature and RH control (65°F / 50% RH), using positively pressurized air from a nearby AHU. Air was distributed to the imaging room via laminar flow diffusers at 25–35 CFM/ft^2, with low-level return placement to ensure directional airflow and effective removal of stale air.
    - Developed a detailed HVAC control sequence, points list, and schematic, incorporating a HEPA-filtered return fan for plenum recirculation. This maintained cleanliness and pressurization in the OR-classified zone while allowing energy-efficient recirculation of filtered return air. All systems were modeled in Revit to ensure 3D coordination, reduce clashes, and maintain alignment with architectural and structural disciplines.

6. Dragon Clean Room Outside Air and Exhaust System DOAS ERU design in Revit (SpaceX)
    - Performed on-site field verification using laser measurement to document existing conditions; developed a complete 3D Revit model of the facility and adjacent spaces, including architectural envelope, ceilings, partitions, and openings, to support accurate HVAC system design.
    - Engineered the outside air and exhaust system based on oxygen-deficiency hazard (ODH) analysis, calculating required exhaust and makeup airflows for a sudden nitrogen release scenario to restore oxygen concentrations to OSHA-compliant levels.

- Designed a dedicated DOAS ERU solution to deliver conditioned outside air and maintain life-safety compliance while integrating with clean room operational requirements.
- Selected and configured a Greenheck DOAS ERU using eCAPS, defining airflow rates, external static pressure, entering air conditions (ASHRAE climate data), space temp and RH targets, and duct and control preferences.
- Delivered a full demolition and new-construction Revit drawing set — from field data to permit-ready documentation — in under one week.

7. HT51 Hydronic VPF heat pump plant and VAV AHU design in Revit (SpaceX)
    - Performed on-site field verification using laser measurement to capture existing conditions; developed a complete and accurate 3D Revit building model from field markups, including architectural envelope, ceilings, partitions, and adjacent spaces to support HVAC and hydronic system design.
    - Engineered three variable-primary-flow (VPF) heat pump plant systems, performing detailed pump head loss calculations to size primary pumps and identify critical hydraulic paths. Plants 1 and 2 were identical, while Plant 3 included fan coil units, resulting in a higher critical path and increased pump head requirements.
    - Designed VPF control strategies utilizing differential pressure sensors at the hydraulically most remote locations to regulate pump speed and maintain minimum flow through bypass circuits under part-load conditions.
    - Developed comprehensive mechanical plans for ductwork, hydronic piping, and equipment layouts, along with detailed controls schematics and sequences of operation for each plant configuration.
    - Engineered a dual-loop hydronic configuration for Plant 3, incorporating both chilled water and heating hot water systems; designed a non-proprietary valve and piping arrangement enabling heat pump chillers to operate in cooling or heating mode independently, allowing simultaneous heating and cooling operation across the plant.
    - Led field verification efforts to confirm mechanical and controls components were installed per design intent, validating system operation and control logic during commissioning readiness reviews.

High-Performance ASHRAE G-36/90.1/105/228 HVAC Sequence of Operation (SOO) Articles:

1. Variable Primary Flow Heat Recovery Chilled Water Plant with CHW/HW/CW loops, HX, and TES SOO
2. Variable Air Volume Dedicated Outside Air Energy Recovery Unit - Variable Relief Pressurization Control SOO
3. Variable Air Volume Series Heat Recovery CHW Air Handling Unit SOO
4. Active Desiccant Dedicated Outside Air Energy Recovery Unit with Heat Pump HW Regeneration and TES SOO
5. Variable Air Volume Terminal Unit with HW Reheat SOO